THE NAIVE COTANGENT COMPLEX OF AN ALGEBRAIC SPACE AGREES WITH THAT OF AN ETALE SCHEME CHART

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5323-5334 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 5336-5395, the complete author proof (58 body lines).

Editorial note. Navigation stubs: 9 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The naive cotangent complex of a morphism, etale localization of differentials, and etale charts for algebraic spaces; the comparison is retained.

Source reference. Original lemma more-algebra-lemma-henselization-NL: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5395–5395. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma more-morphisms-definition-netherlander: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5395–5395. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma sites-lemma-point-morphism-sites: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5395–5395. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma sites-modules-lemma-pullback-NL: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5395–5395. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma sites-modules-lemma-pullback-stalk: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5395–5395. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-perfect-remark-match-total-direct-images: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5395–5395. Cited by the retained statement or proof but not part of the collected spans;

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supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-perfect-section-derived-quasi-coherent-etale: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5395–5395. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-properties-lemma-describe-etale-local-ring: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5395–5395. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-properties-theorem-exactness-stalks: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5395–5395. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume X and Y representable by schemes X_0 and Y_0. Then there is a canonical identification $NL_{X/Y} = \epsilon^* NL_{X_0/Y_0}$ in $D(\mathcal{O}_X)$ where ϵ is as in Derived Categories of Spaces, Section spaces-perfect-section-derived-quasi-coherent-etale and NL_{X_0/Y_0} is as in More on Morphisms, Definition more-morphisms-definition-netherlander.*

Proof. Let $f_0 : X_0 \to Y_0$ be the morphism of schemes corresponding to f. There is a canonical map $\epsilon^{-1} f_0^{-1} \mathcal{O}_{Y_0} \to f_{small}^{-1} \mathcal{O}_Y$ compatible with $\epsilon^\sharp : \epsilon^{-1} \mathcal{O}_{X_0} \to \mathcal{O}_X$ because there is a commutative diagram

$$\begin{array}{ccc} X_{0,Zar} & \xleftarrow{\epsilon} & X_{étale} \\ {\scriptstyle f_0}\downarrow & & \downarrow{\scriptstyle f} \\ Y_{0,Zar} & \xleftarrow{\epsilon} & Y_{étale} \end{array}$$

see Derived Categories of Spaces, Remark spaces-perfect-remark-match-total-direct-images. Thus we obtain a canonical map

$$\epsilon^{-1} NL_{X_0/Y_0} = \epsilon^{-1} NL_{\mathcal{O}_{X_0}/f_0^{-1}\mathcal{O}_{Y_0}} = NL_{\epsilon^{-1}\mathcal{O}_{X_0}/\epsilon^{-1}f_0^{-1}\mathcal{O}_{Y_0}} \to NL_{\mathcal{O}_X/f_{small}^{-1}\mathcal{O}_Y} = NL_{X/Y}$$

by functoriality of the naive cotangent complex. To see that the induced map $\epsilon^* NL_{X_0/Y_0} \to NL_{X/Y}$ is an isomorphism in $D(\mathcal{O}_X)$ we may check on stalks at geometric points (Properties of Spaces, Theorem spaces-properties-theorem-exactness-stalks). Let $\overline{x} : \mathrm{Spec}(k) \to X_0$ be a geometric point lying over $x \in X_0$, with $\overline{y} = f \circ \overline{x}$ lying over $y \in Y_0$. Then

$$NL_{X/Y,\overline{x}} = NL_{\mathcal{O}_{X,\overline{x}}/\mathcal{O}_{Y,\overline{y}}}$$

This is true because taking stalks at $\overline{x}$ is the same as taking inverse image via $\overline{x} : \mathrm{Spec}(k) \to X$ and we may apply Modules on Sites, Lemma sites-modules-lemma-pullback-NL. On the other hand we have

$$(\epsilon^* NL_{X_0/Y_0})_{\overline{x}} = NL_{X_0/Y_0,x} \otimes_{\mathcal{O}_{X_0,x}} \mathcal{O}_{X,\overline{x}} = NL_{\mathcal{O}_{X_0,x}/\mathcal{O}_{Y_0,y}} \otimes_{\mathcal{O}_{X_0,x}} \mathcal{O}_{X,\overline{x}}$$

Some details omitted (hint: use that the stalk of a pullback is the stalk at the image point, see Sites, Lemma sites-lemma-point-morphism-sites, as well as the corresponding result for modules, see Modules on Sites, Lemma sites-modules-lemma-pullback-stalk). Observe that $\mathcal{O}_{X,\overline{x}}$ is the strict henselization of $\mathcal{O}_{X_0,x}$ and

similarly for $\mathcal{O}_{Y,\overline{y}}$ (Properties of Spaces, Lemma spaces-properties-lemma-describe-etale-local-ring). Thus the result follows from More on Algebra, Lemma more-algebra-lemma-henselization-NL. □